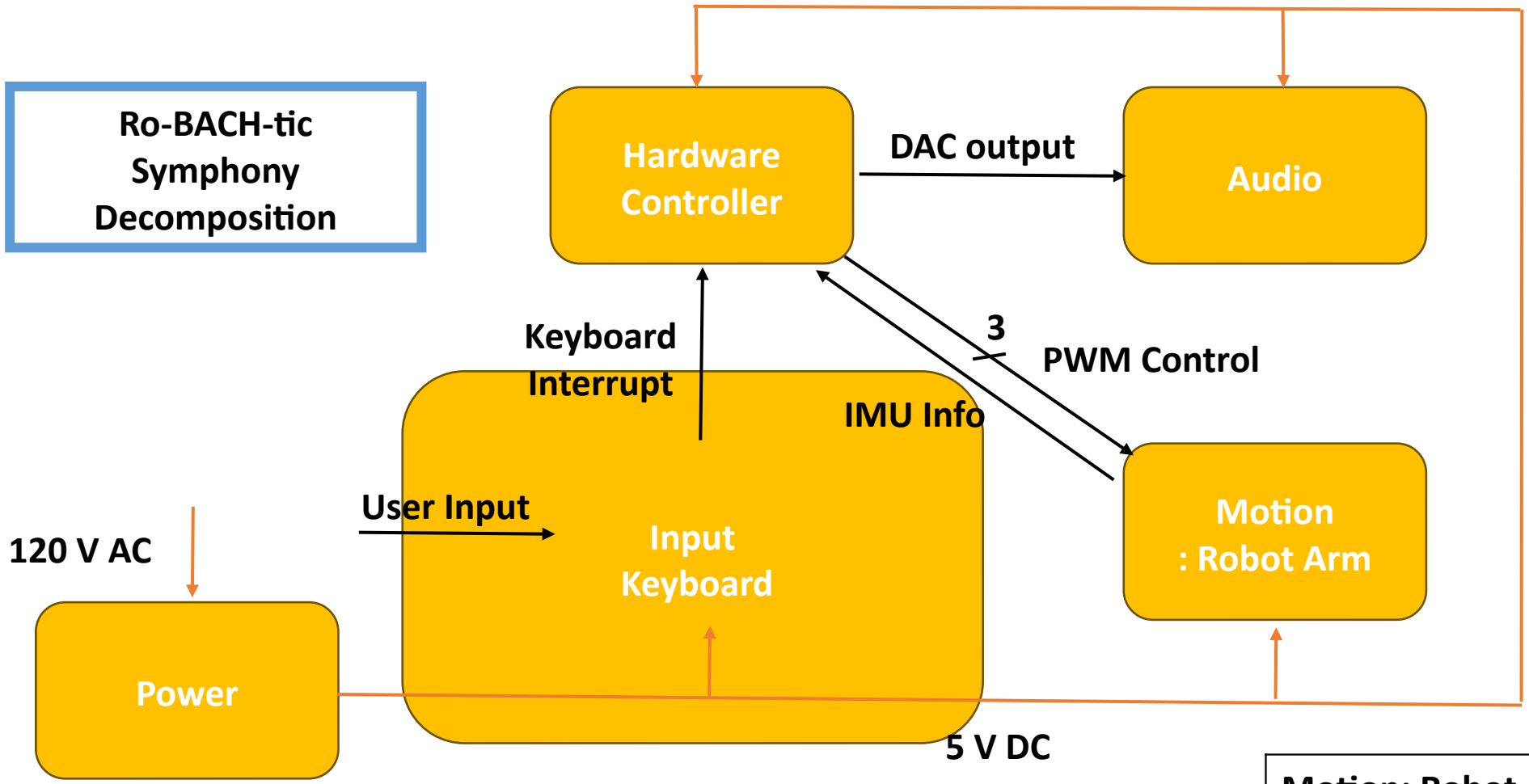
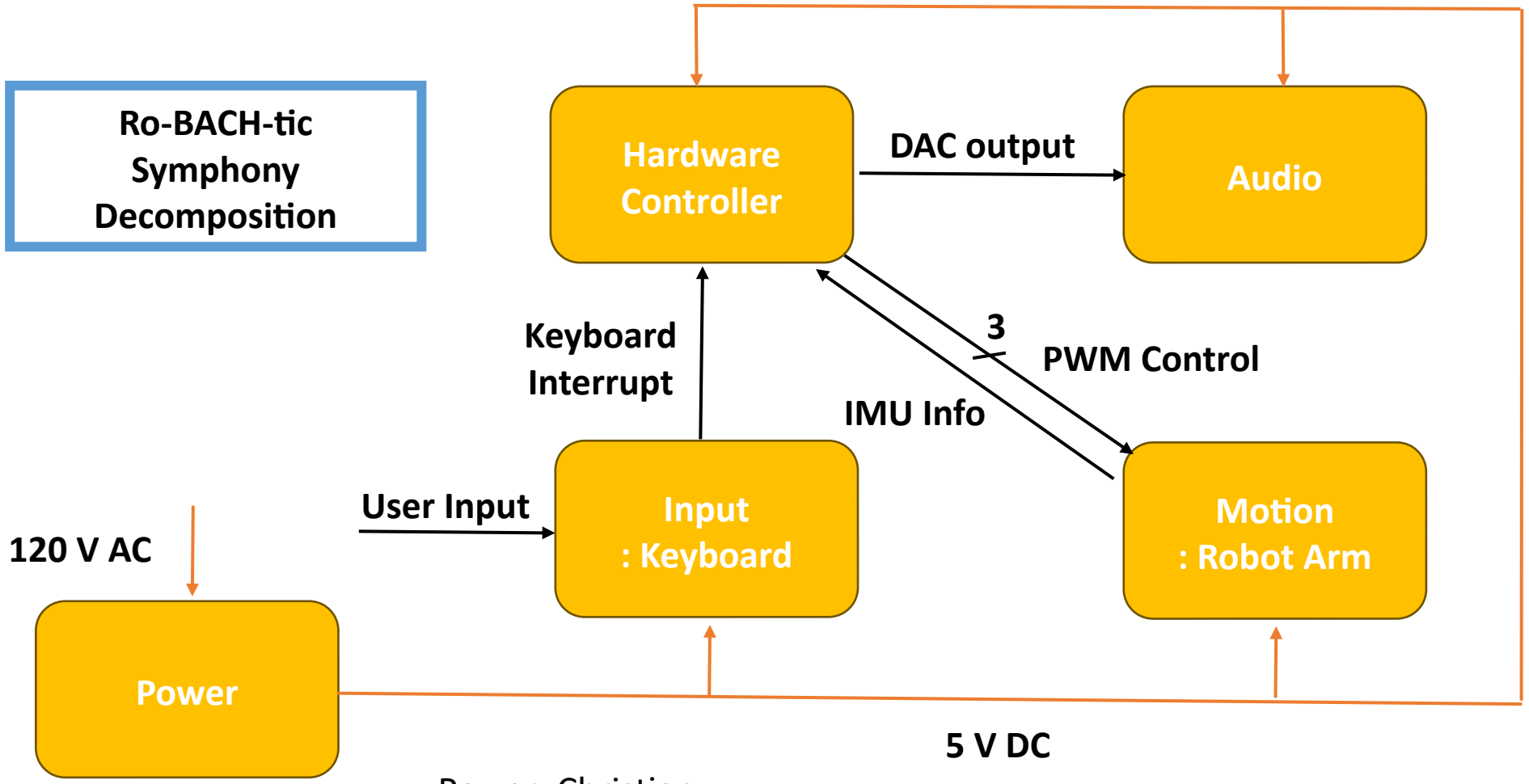


Input: Keyboard
- Keyboard box   • Black Key x5   • White Key Left x2   • White Key Middle x3   • White Key Right x2   • Button Extender x4 - PCB   • Button x16   • 10k Register x19   • GPIO Extender   • GPIO 5 pin

Motion: Robot Arm
- Robot Arm   • Biceps   • Forearm   • Bow   • Bow Gear   • Servo x3 - Violin   • Chin Rest   • Peg   • Neck   • Violin Body - Main Box   • Gear Box   ○ Big Gear   ○ Small Gear   • Main Shaft

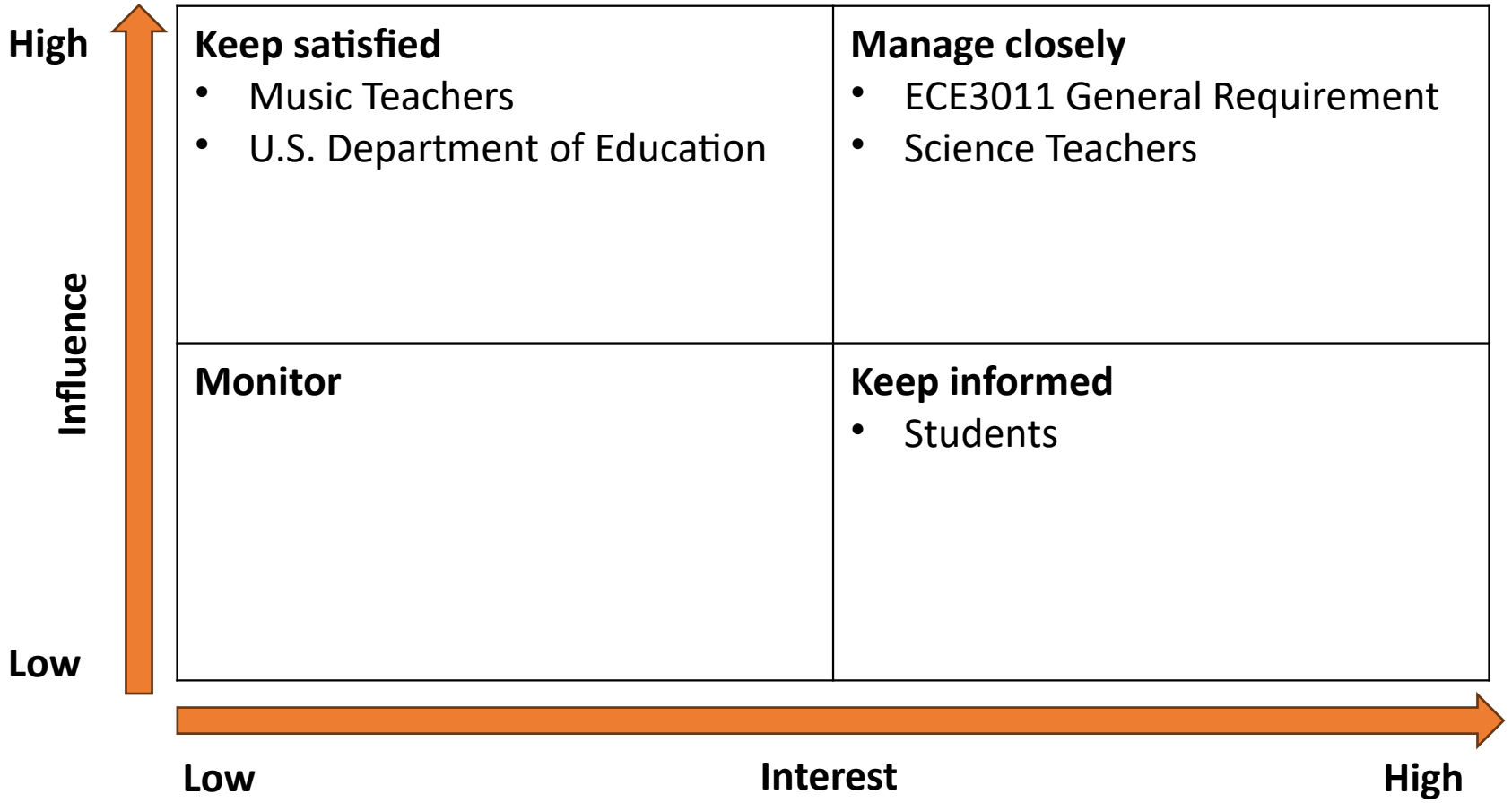
Ro-BACH-tic Symphony Decomposition



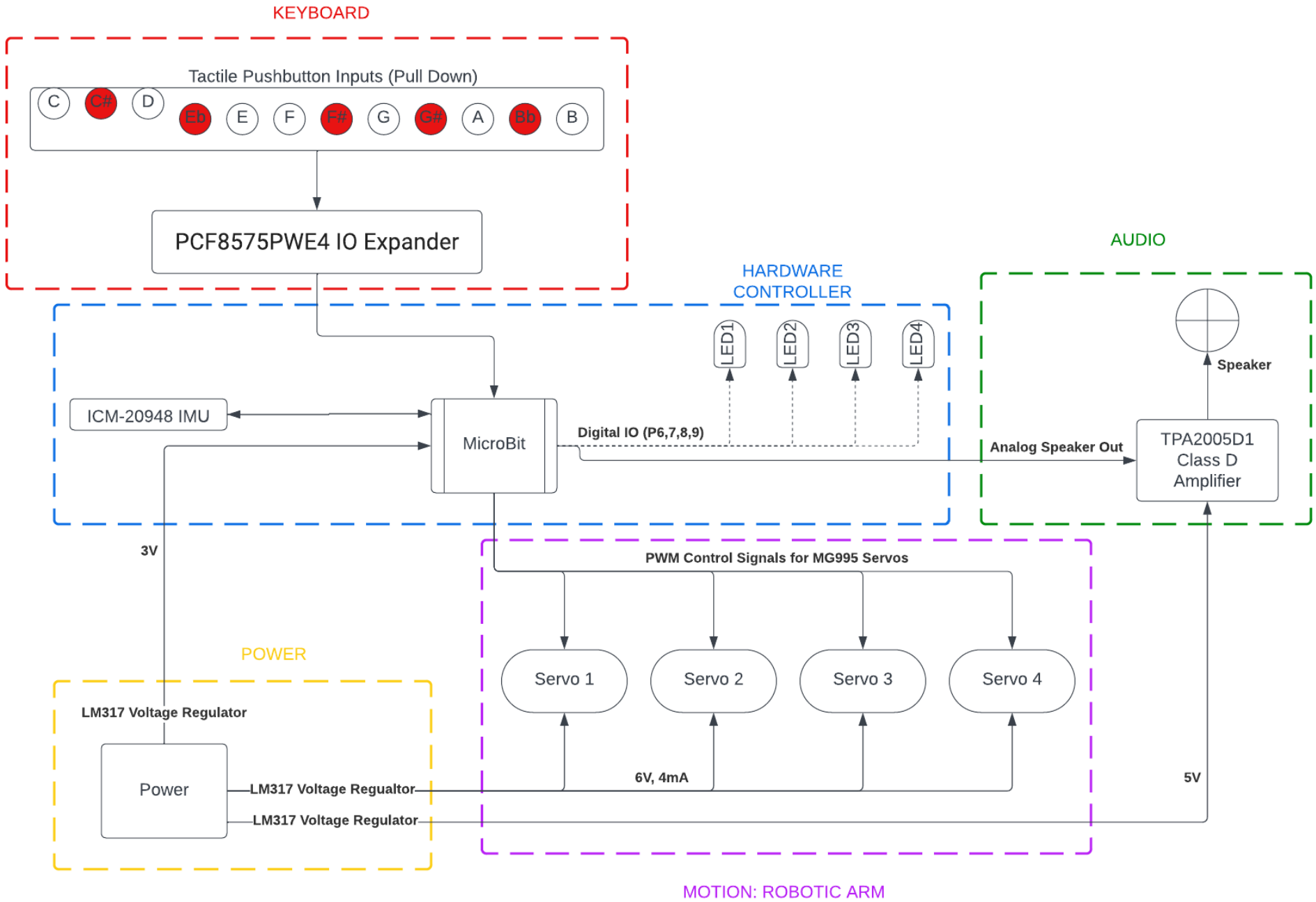
Power: Christian
Audio: Christian

Hardware Controller: Christoper
Input: Christoper

Mechanical: Jeon, Victor



- Science Teachers
- Music Teachers
- Students
- U.S. Department of Education


KEYBOARD
Tactile Pushbutton Inputs (Pull Down)
C
C#
D
Eb
E
F
F#
G
G#
A
Bb
B
PCF8575PWE4 IO Expander
HARDWARE CONTROLLER
LED1
LED2
LED3
LED4
ICM-20948 IMU
MicroBit
Digital IO (P6,7,8,9)
AUDIO
Speaker
Analog Speaker Out
TPA2005D1 Class D Amplifier
3V
PWM Control Signals for MG995 Servos
POWER
Servo 1
Servo 2
Servo 3
Servo 4
LM317 Voltage Regulator
Power
LM317 Voltage Regualtor
LM317 Voltage Regulator
6V, 4mA
5V
MOTION: ROBOTIC ARM